

Introduction

This project investigates neural signatures associated with inner speech and examines whether phonemic information can be decoded directly from EEG recordings. Using a publicly available speech-related EEG dataset and computational resources, we developed a customized processing pipeline that combines task-specific signal preparation, visual analyses of spatiotemporal EEG dynamics, and a machine learning based phoneme classification framework optimized for decoding performance.

Through this work, we sought to assess how discriminative phonetic information is represented in non-invasive electrophysiological signals and to establish a practical foundation for EEG-driven speech decoding brain computer interfaces (BCIs) capable of translating covert speech-related activity into linguistic information.

1. EEG Dataset & Experimental Paradigm

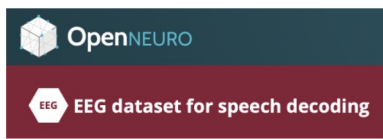
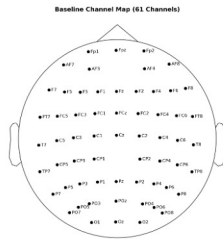


Figure 1A. Channel Map



Dataset & Experimental Paradigm

- 2021 cohort: 16 participants, 64-channel EEG, sampled at 2,000 Hz
- Single phoneme: listen & repeat
- Phoneme pairs (CV): identify the consonant
- Words (CVC): classify real vs. pseudoword
- TMS: paired pulses (50-ms interval), followed by auditory stimulus after 50 ms

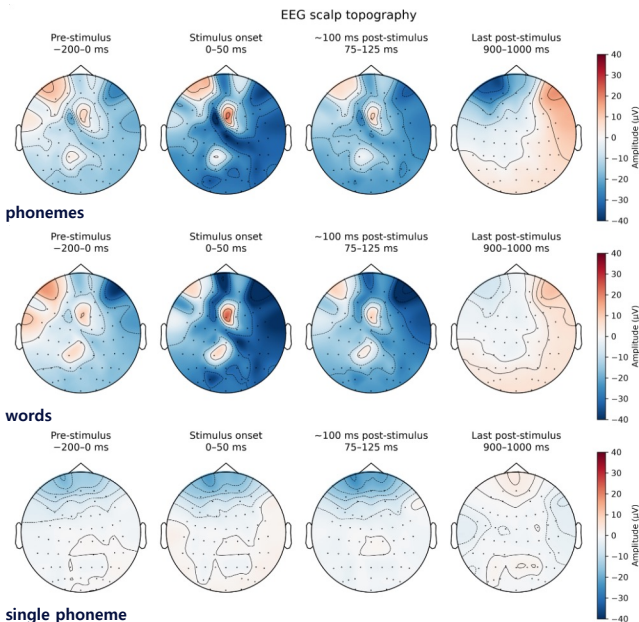
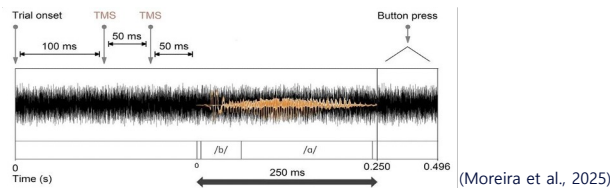


Figure 1B. Task-Specific Spatiotemporal EEG Topography

2. EEG Preprocessing

Objective

To remove noise and artifacts from raw EEGLAB recordings and transform the EEG signals into a standardized format suitable for machine-learning analysis.

Preprocessing Output

The cleaned EEG data were segmented into 1-s epochs and organized into a model-ready 3D tensor:

$$X = [\text{Trials} \times 61 \text{ Channels} \times 256 \text{ Samples}]$$

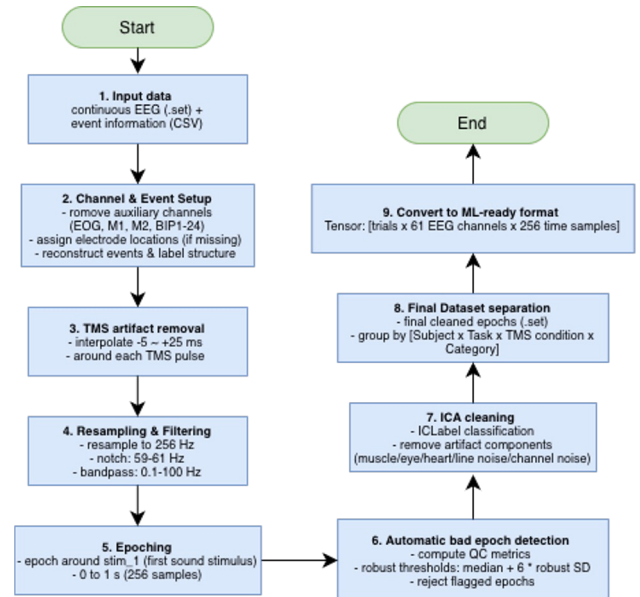


Figure 2A. Preprocessing Flowchart

Data Quality Validation

Epoch-level QC: Robust signal-quality metrics were used to detect and reject contaminated trials automatically.

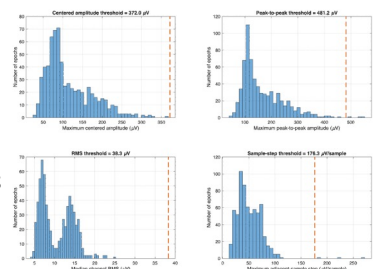


Figure 2B. Automatic Bad-Epoch Rejection Thresholds

ICA-level QC: ICLabel-based classification was used to identify and remove residual non-neural components.



Neural Speech Decoding for Brain Computer Interfaces

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3. Stimulus-Locked EEG Signal Characterization

Data

- Dataset: S02_3_filtered_data.set/.fdt
- Event reference: stim_1 = 0 s

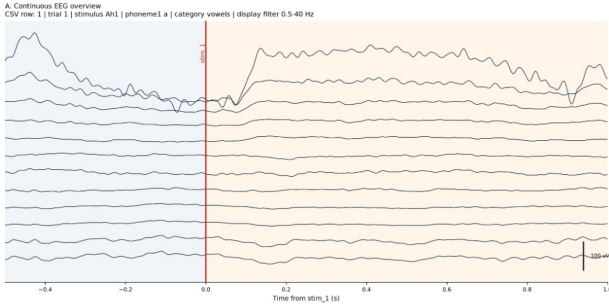


Figure 3A. Continuous EEG & Event Alignment

- Single-trial EEG aligned to stim_1 (t = 0 s).
- Confirms temporal synchronization between continuous EEG and auditory stimulus onset

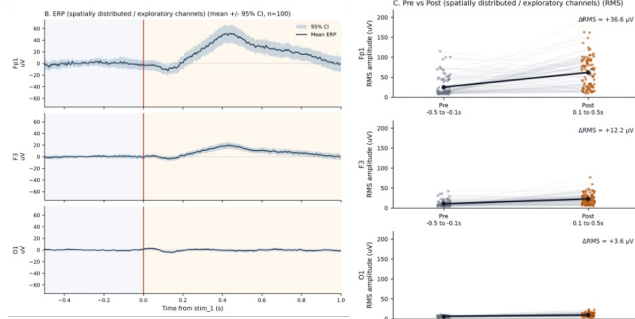


Figure 3B. Averaged ERP

- Trials were aligned to stim_1 and averaged.
- A small N100-like negative deflection was followed by a late positive response, peaking at approximately 400–500 ms.
- The response was stronger over frontal/fronto-central electrodes and weaker at O1

Figure 3C. Pre-Post Stimulus RMS

- RMS compared between:
 - Pre: -0.5 to -0.1 s
 - Post: 0.1 to 0.5 s
- RMS amplitude increased after stimulus onset, particularly at frontal electrodes.

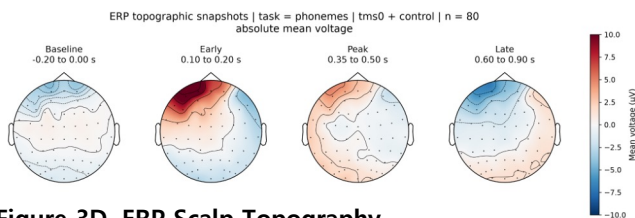


Figure 3D. ERP Scalp Topography

- Stimulus-related activity showed a clear time-dependent spatial redistribution.
- Temporal changes across 100–500 ms are consistent with dynamic phonetic-feature processing reported for speech stimuli

4. Machine-Learning-Based Model

Notes

- Our goal was not to build a practically deployable model.
- Rather than prioritizing model speed, our goal was to incorporate into the model architecture the factors that we believe the model should consider, using mathematical formulations and statistical computations.

Goals

- Subject-specific model: Different LDA for each subject.
- 11-phoneme classification: Classify 11 phonemes (a, i, e, u, o, s, z, p, b, t, d) — chance level: 0.091.
- Input: EEG signals that were recorded while the subject listened to the phonemes.
- Output: One of the 11 phonemes.

Model

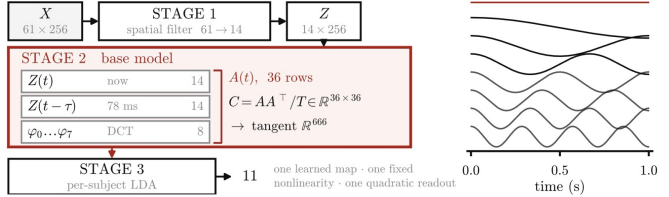


Figure 4A. (left) Model Architecture. (right) DCT Temporal Basis.

- Stage 1:** Noise removal and dimensionality reduction in order to ensure stability
 - $Z = WT^T X$
 - Maximize the signal differences between phonemes
- Stage 2:** Base model; extract information along the temporal and spatial dimensions
 - Delay embedding ($\tau=20, 78$ ms)
 - Capture temporal changes (dynamics)
 - DCT temporal basis ($\varphi_0 \dots \varphi_7$)
 - Preserve “when” within the representation
 - Riemannian geometry (log-Euclidean)
 - Flatten matrices on a curved manifold into vectors
- Stage 3:** Extract subject-specific characteristics

Result

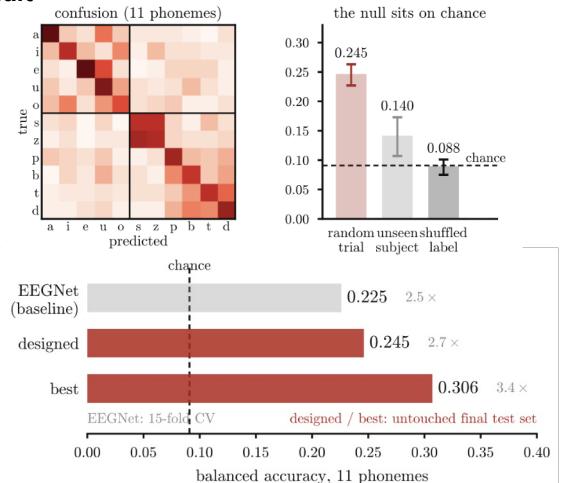


Figure 4B. (top left): Confusion Matrix. (top right): Accuracy by Trial.

- Random Trial: Phoneme classification accuracy on trials that were not included in the training data.
- Unseen Subject: Phoneme classification accuracy for a new subject who was not included in the training data.
- Shuffled Label: Classification accuracy when the labels were randomly shuffled.
- (bottom): Comparison of 11-phoneme classification accuracy among EEGNet, our model (designed), and the additional experimental model (best).